

Emergency Shutdown & Temperature Stabilization Protocol for Diesel Hydrotreater Reactor Feed Surge Drum (D-317)

Document ID	SOP-DH-527-REV1
Facility	Area 300 – Diesel Hydrotreater & Hydrogen Recovery Unit (Western Process Complex)
Effective Date	September 2, 2024
Compliance	OISD-STD-105, API 521, ISO 45001:2018, PESO Static Pressure Vessels Rules 2018

Purpose

Standard operating procedure covering emergency shutdown, stabilization, LOTO, emergency blowdown and RCA for Reactor Feed Surge Drum D-317.

Equipment Specifications

Parameter	Value
Primary Asset	D-317 – Reactor Feed Surge Drum
Associated Pump	P-317B – Reactor Feed Pump B
Normal Pressure	36.8 bar
MAWP	42.5 bar
Temperature	345°C
High Temp Alarm	355°C
High Pressure Trip	40.8 bar
PSV Setpoint	41.8 bar
Emergency Blowdown Valve	BDV-317

Emergency Shutdown Procedure

- Notify CCR, reduce feed, stop Pump P-317B and isolate FCV-317.
- Purge with nitrogen at 1.5–2.5 barg for 5 minutes, vent to flare until oxygen is below 2% and hydrocarbons below 10% LEL.
- Repeat purge twice and maintain a 0.3 barg nitrogen blanket.
- Monitor pressure, shell temperature and hydrogen flow while limiting cooling to 30°C/hour.

LOTO

- Lock electrical isolator for Pump P-317B.
- Close D-317-IN-05 and apply lock/tag.
- Close D-317-OUT-06 and apply lock/tag.
- Drain, vent and verify zero pressure before maintenance.

Root Cause Analysis

Incident ID: INC-2024-143

Event Date: August 19, 2024

Failure Summary: Excessive pump vibration resulted in an automatic trip and transient pressure rise within D-317.

Root Cause: Partial blockage of lubricating oil filter LO-317 reduced bearing lubrication, causing overheating and vibration.

CAPA: Replace bearings and filter, flush lubrication circuit, install online oil particle counter, implement DCS trend analytics, shorten oil analysis intervals and enhance predictive maintenance.